

Mechanistic Interpretability and Quantization Analysis - Progress Report

Introduction

This report summarizes my understanding of a mechanistic interpretability project focused on analyzing how quantization affects transformer representations.

Problem Statement

The goal is to study not only performance degradation after quantization but also the internal reasons for that degradation.

Pipeline Overview

A frozen pretrained language model is used. Hidden activations are extracted from intermediate transformer layers and analyzed.

Sparse Autoencoders

Sparse Autoencoders learn compact and interpretable features from transformer activations and can reveal meaningful internal structures.

Quantization Analysis

Experiments compare 8-bit, 4-bit, and 2-bit quantization using metrics such as Mean Squared Error and similarity measures.

Representation Damage

Activation differences between original and quantized representations are measured to identify sensitive neurons and features.

Neuron Ablation

Important neurons can be identified by selectively disabling them and observing the effect on model behavior.

Visualization

Methods such as t-SNE and UMAP can visualize how hidden representations change after quantization.

Expected Deliverables

Activation extraction, quantization experiments, similarity metrics, visualization, neuron sensitivity analysis, and conclusions.

Learning Outcomes

Studying transformers and attention mechanisms provided the foundation required to understand mechanistic interpretability workflows.

Future Work

Future work includes implementing hidden-state extraction, training Sparse Autoencoders, and performing systematic quantization studies.